

Industrial Internship Report on "Quality Prediction in Mining Process using Machine Learning"

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Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus and the IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT).

This internship was focused on a project/problem statement provided by UCT. We were required to complete the project, including its implementation, documentation, and report, within a duration of 6 weeks.

My project was "**Quality Prediction in Mining Process using Machine Learning.**" The objective of this project was to develop a machine learning-based predictive model to estimate the percentage of silica concentrate in a mining flotation process. The project involved data preprocessing, exploratory data analysis, feature engineering, correlation analysis, model development using the Random Forest Regression algorithm, model evaluation, feature importance analysis, and prediction visualization using Python libraries such as Pandas, NumPy, Matplotlib, Scikit-learn, and Joblib.

This internship provided me with valuable exposure to real-world industrial data and machine learning applications. It enhanced my understanding of data analysis, predictive modeling, and model deployment while improving my technical and problem-solving skills. Overall, it was a highly enriching learning experience.

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1 Preface

This report presents the work completed during my Industrial Internship conducted by Upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship provided an opportunity to work on a real-world machine learning project titled **"Quality Prediction in Mining Process using Machine Learning."**

During this six-week internship, I gained practical experience in applying machine learning techniques to solve an industrial mining process problem. The project involved understanding the mining flotation dataset, performing data preprocessing, conducting exploratory data analysis, developing a predictive model using Random Forest Regression, evaluating the model using performance metrics, and analyzing feature importance to identify the factors affecting silica concentrate quality.

Throughout the internship, I enhanced my knowledge of Python, data preprocessing, feature engineering, exploratory data analysis (EDA), model evaluation, visualization, and machine learning using libraries such as Pandas, NumPy, Matplotlib, Scikit-learn, and Joblib. I also learned how to organize a machine learning project, document it professionally, save trained models, and maintain the project using Git and GitHub.

This internship strengthened my practical knowledge of machine learning and data analytics while improving my problem-solving, analytical, and technical documentation skills. It also provided valuable exposure to solving real-world industrial problems using artificial intelligence and predictive analytics. I sincerely thank **Upskill Campus, UniConverge Technologies Pvt. Ltd. (UCT)**, my college, and all the mentors who guided and supported me throughout this internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.



i. UCT IoT Platform (**uct Insight**)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY **WATCH**

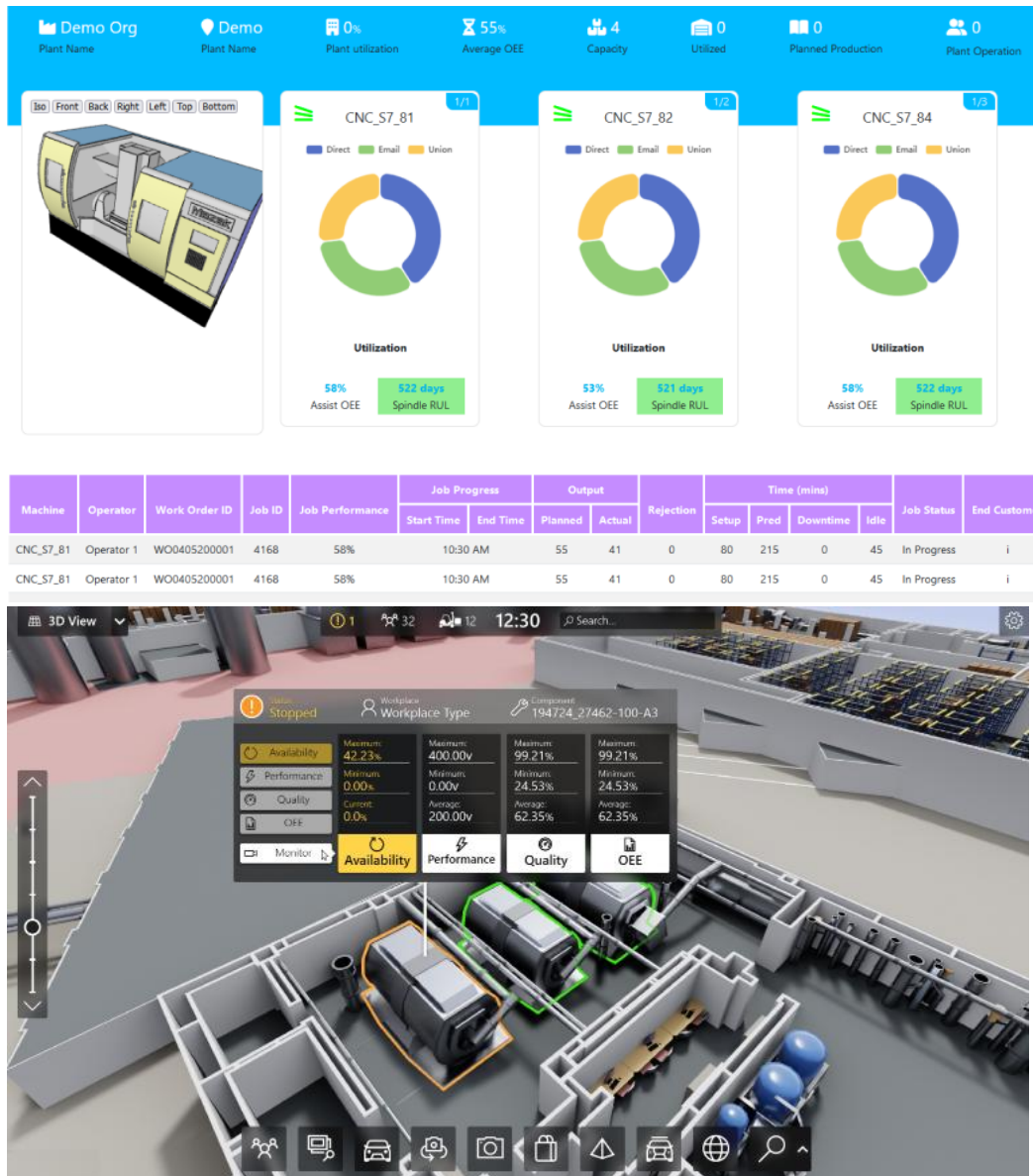
ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



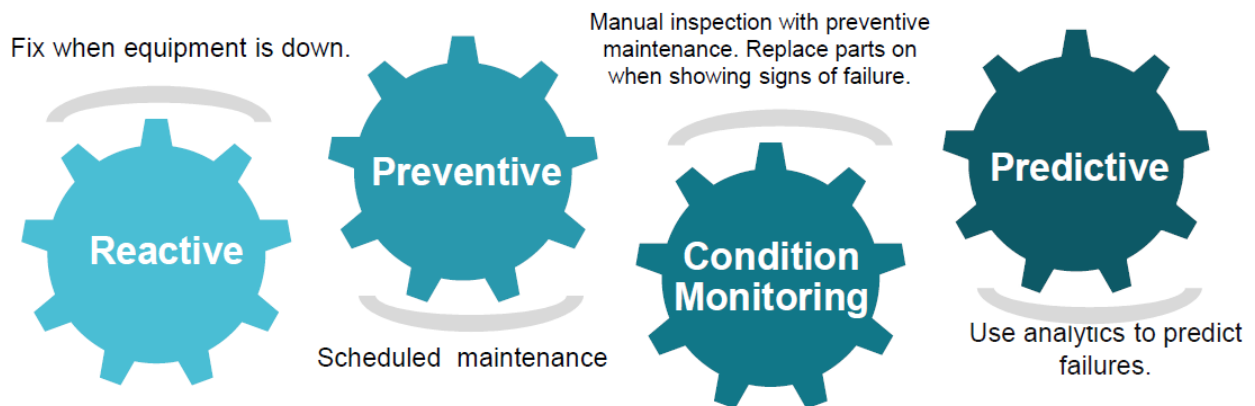


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

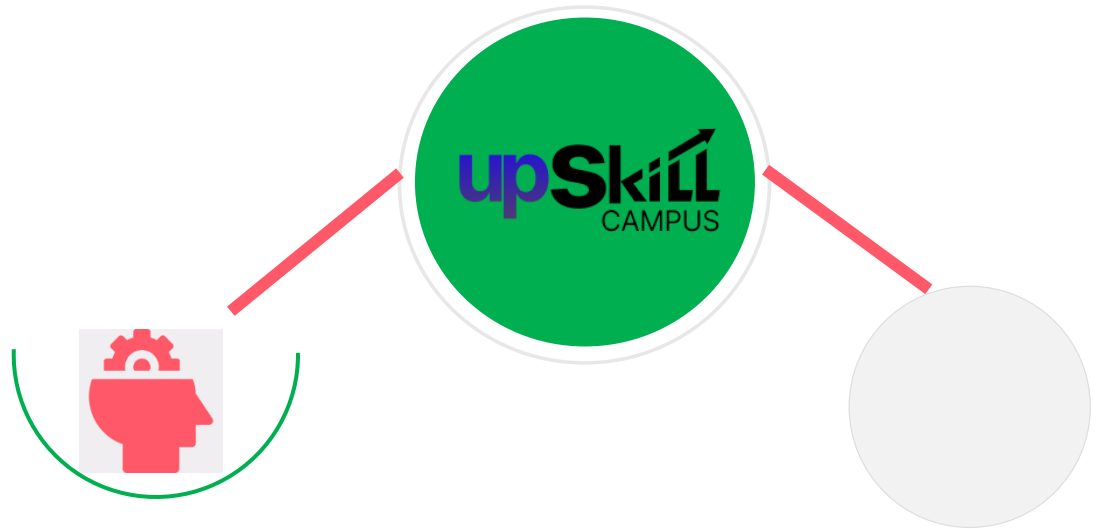
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

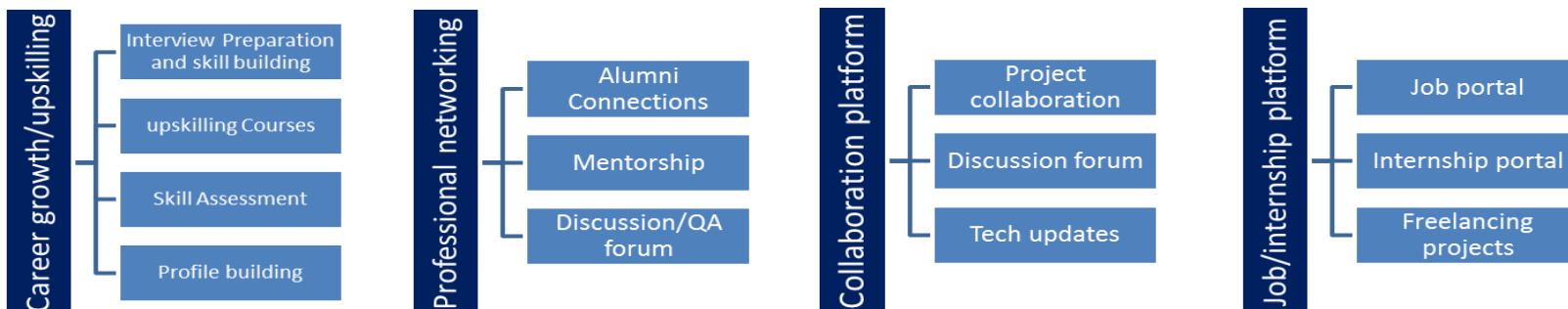
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ▣ get practical experience of working in the industry.
- ▣ to solve real world problems.
- ▣ to have improved job prospects.
- ▣ to have Improved understanding of our field and its applications.
- ▣ to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] **Upskill Campus**, "Industrial Internship Program." Available: <https://www.upskillcampus.com/>
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- [3] **Scikit-learn Developers**, "Scikit-learn Documentation." Available: <https://scikit-learn.org/stable/>
- [4] **Pandas Development Team**, "Pandas Documentation." Available: <https://pandas.pydata.org/docs/>

2.6 Glossary

Terms	Acronym
Machine Learning	ML
Mean Absolute Error	MAE
Random Forest Regressor	RFR
Exploratory Data Analysis	EDA
Root Mean Squared Error	RMSE
Coefficient of Determination	R ²

3 Problem Statement

The assigned problem statement was "**Quality Prediction in Mining Process using Machine Learning.**" The objective of this project was to develop a machine learning model capable of predicting the percentage of silica concentrate produced during the mining flotation process based on various operational parameters.

The project involved analyzing a real-world mining dataset containing features such as iron feed percentage, silica feed percentage, starch flow, amina flow, ore pulp flow, pulp pH, pulp density, air flow, and flotation column levels. The data was preprocessed to handle missing values and prepare it for analysis. Exploratory Data Analysis (EDA) and feature engineering were performed to understand relationships between variables and identify the most influential features.

A **Random Forest Regression** model was developed to predict the silica concentrate percentage. The model was evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score, demonstrating high prediction accuracy. The project helps improve process monitoring, supports data-driven decision-making, and enhances the efficiency and quality of mineral processing operations.

4 Existing and Proposed solution

4.1.1 Existing Solutions and Their Limitations

Existing solutions for predicting mining process quality mainly rely on manual monitoring, statistical analysis, and traditional machine learning models such as Linear Regression and Decision Tree Regression. These approaches often fail to capture complex relationships among process parameters, resulting in lower prediction accuracy. Additionally, manual monitoring is time-consuming, prone to human error, and unsuitable for real-time quality prediction.

4.1.2 Proposed Solution

The proposed solution is a Random Forest Regression-based machine learning model that predicts the percentage of silica concentrate using process parameters such as iron feed, silica feed, pulp flow, pH, density, air flow, and flotation column levels. The model includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and visualization to achieve accurate quality prediction.

4.1.3 Value Addition

- Improves prediction accuracy compared to traditional approaches.
- Reduces manual effort and human errors in quality monitoring.
- Identifies the most important process parameters affecting product quality.
- Enables data-driven decision-making for process optimization.
- Supports improved efficiency and consistency in mining operations through predictive analytics.

4.2 Code submission (Github link)

<https://github.com/anujaghodele22-dev/Quality-Prediction-in-Mining-Process>

4.3 Report submission (Github link)

<https://github.com/anujaghodele22-dev/Quality-Prediction-in-Mining-Process>

5 Proposed Design/ Model

The proposed system is a machine learning-based predictive model developed to estimate the percentage of silica concentrate in a mining flotation process. The system follows a structured workflow beginning with data collection and preprocessing, followed by exploratory data analysis, feature engineering, model training, evaluation, and prediction.

Initially, the mining process dataset is loaded and preprocessed to remove missing values and prepare the data for analysis. Exploratory Data Analysis (EDA) is performed to understand the distribution of features and their relationships. Correlation analysis is used to identify important parameters affecting the output. The dataset is then divided into training and testing sets.

A Random Forest Regression model is trained using the processed data. The trained model predicts the silica concentrate percentage and is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Finally, the trained model is saved using Joblib for future predictions.

5.1 High Level Diagram (if applicable)

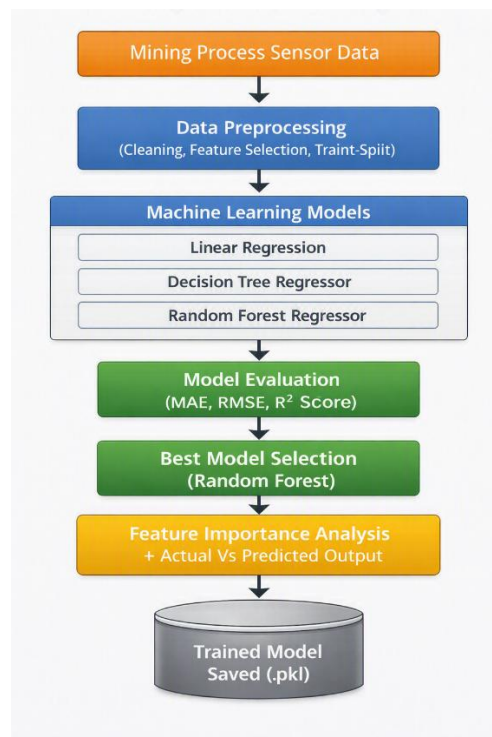


Figure 1: Quality Prediction in Mining Process using Machine Learning

5.2 Low Level Diagram (if applicable)

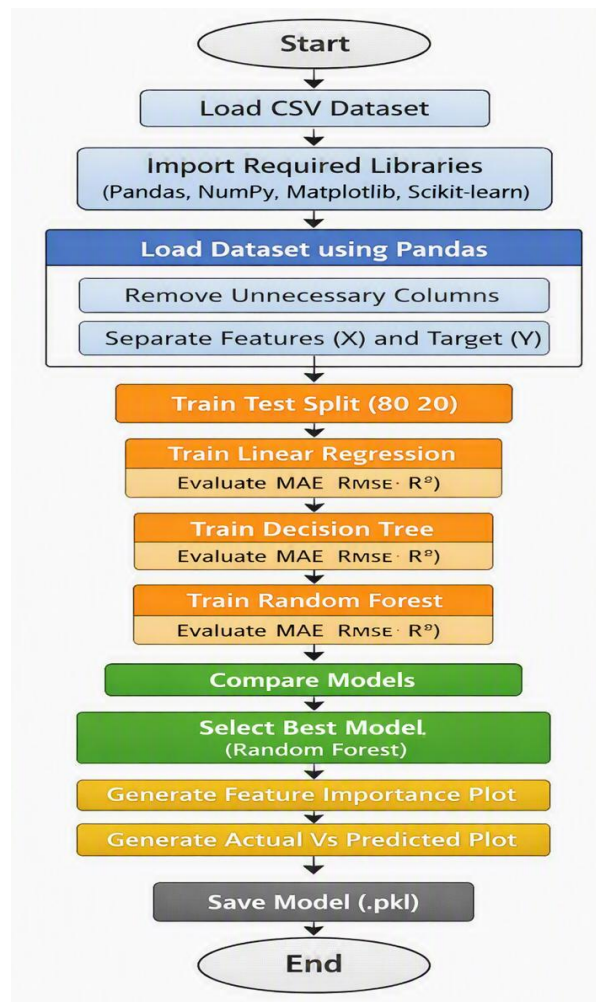


Figure 5.2: Low-Level Design of Manufacturing Process Prediction System

5.3 Interfaces (if applicable)

The project was implemented using Jupyter Notebook as the development environment. The system interface allows the user to load the mining flotation dataset, perform data preprocessing, exploratory data analysis (EDA), feature engineering, and train the Random Forest Regression model

using Python code cells. The results are presented through evaluation metrics (MAE, RMSE, and R^2 Score), data tables, and graphical visualizations such as the Distribution of Silica Concentrate, Correlation Matrix, Feature Importance Plot, and Actual vs Predicted Output Plot. Finally, the trained model is saved as a .pkl file using Joblib, enabling the model to be reused for future quality predictions without retraining.

This version is specific to your Quality Prediction in Mining Process using Machine Learning project and matches the work you completed.

6 Performance Test

The proposed solution was designed to predict the quality of the mining flotation process with high accuracy while handling a large industrial dataset. The main constraints identified were large dataset size, model training time, and prediction accuracy.

To address these challenges, data preprocessing and feature selection were performed to improve data quality and reduce unnecessary computations. The Random Forest Regressor was selected because it provides high prediction accuracy and handles complex relationships between process parameters effectively.

The model was evaluated using MAE, RMSE, and R^2 Score. The obtained R^2 Score of 0.9982 indicates excellent prediction accuracy, demonstrating that the proposed solution is suitable for industrial quality prediction.

Future improvements may include optimizing the model for faster training and deploying it for real-time quality prediction in mining plants.

6.1 Test Plan/ Test Cases

Objective:

To evaluate the performance of the proposed Random Forest Regression model for predicting the percentage of silica concentrate in the mining flotation process.

Test Case ID	Description	Expected Result	Status
TC-01	Load the mining flotation dataset	Dataset loads successfully without errors	Passed
TC-02	Preprocess the dataset	Data is cleaned and prepared for model	Passed

Test Case ID	Description	Expected Result	Status
		training	
TC-03	Perform Exploratory Data Analysis (EDA)	Data distribution and correlations are generated successfully	Passed
TC-04	Train the Random Forest Regression model	Model trains successfully	Passed
TC-05	Evaluate the model using MAE, RMSE, and R ² Score	Performance metrics are generated correctly	Passed
TC-06	Generate Feature Importance and Actual vs Predicted plots	Visualizations are generated successfully	Passed
TC-07	Save the trained model using Joblib	Model is saved successfully as random_forest_model.pkl	Passed

6.2 Test Procedure

The Mining Process Flotation Plant dataset was loaded into Jupyter Notebook using the Pandas library. The dataset was preprocessed by handling missing values and preparing the data for analysis. Exploratory Data Analysis (EDA) and correlation analysis were performed to understand the relationships among the process parameters.

The dataset was divided into training and testing sets using an 80:20 ratio. A Random Forest Regression model was trained using the training dataset and evaluated on the testing dataset. The model's performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R² Score. Feature Importance and Actual vs Predicted plots were generated to analyze the model's prediction performance. Finally, the trained model was saved as `random_forest_model.pkl` using Joblib for future predictions.

6.3 Performance Outcome

Model	MAE	RMSE	R ² Score
Linear Regression	0.4921	0.6364	0.6795
Decision Tree Regressor	0.2953	0.4305	0.8534

Model	MAE	RMSE	R ² Score
Random Forest Regressor	0.0104	0.0471	0.9982

The performance of three regression models was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Among all the models, the Random Forest Regressor achieved the best results with the lowest MAE (0.0104), lowest RMSE (0.0471), and the highest R^2 Score (0.9982), indicating excellent prediction accuracy. The Decision Tree Regressor performed better than Linear Regression, but its accuracy was lower than that of the Random Forest model. Therefore, the Random Forest Regressor was selected as the final model for predicting the % Silica Concentrate in the mining process.

7 My learnings

During this internship, I gained practical experience in developing a complete machine learning solution for predicting the % Silica Concentrate in a mining flotation process. I improved my understanding of Python programming and learned how to use libraries such as Pandas, NumPy, Matplotlib, Scikit-learn, and Joblib for data preprocessing, model development, visualization, and model deployment.

I learned how to:

- Perform data preprocessing and handle missing values.
- Analyze and visualize mining process data.
- Split the dataset into training and testing sets.
- Train and evaluate Linear Regression, Decision Tree Regressor, and Random Forest Regressor models.
- Compare model performance using MAE, RMSE, and R^2 Score.
- Analyze feature importance using the Random Forest model.
- Visualize prediction results using Actual vs Predicted plots and Feature Importance graphs.
- Save the trained machine learning model using Joblib for future predictions.
- Manage the project using GitHub and prepare professional technical documentation.

This internship enhanced my problem-solving, data analysis, and machine learning skills while providing valuable exposure to solving real-world industrial problems in the mining and mineral processing domain.

8 Future work scope

The current project focuses on predicting the % Silica Concentrate in a mining flotation process using machine learning regression models. In the future, the project can be enhanced by:

- Using larger and more diverse mining process datasets to improve model generalization.
- Applying advanced machine learning algorithms such as XGBoost, LightGBM, and CatBoost for higher prediction accuracy.
- Performing hyperparameter tuning and cross-validation to optimize model performance.
- Developing a web application using Flask or Streamlit for real-time quality prediction.
- Integrating the trained model with IoT sensors for continuous monitoring of mining process parameters.
- Deploying the model on cloud platforms for industrial-scale applications.
- Implementing real-time dashboards to monitor process quality, predictions, and key performance indicators.

These enhancements can improve the accuracy, scalability, and practical usability of the system for real-world mining and mineral processing industries.

